

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Game Based Learning Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 40 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A Game Based Learning Assessment

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Understanding of the Problem / Game Objective	10%	
Application of AI Search / Problem-Solving Technique	15%	
Successful Completion of the Game / Puzzle	20%	
Efficiency of Solution / Optimal Moves	10%	
Documentation / Evidence of Completion	15%	
Understanding of the Problem / Game Objective	15%	
Originality and Critical Thinking	15%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

Answer All the Questions

Q. No.	Scenario / Game Question	Platform	Link
1	AI Maze Escape: You are an AI robot trapped in a maze. Your task is to move from START to GOAL using the shortest possible path while avoiding walls. Challenge: Find the shortest path and record the number of steps taken.	Scratch	https://scratch.mit.edu
2	N-Queens Challenge: You are an AI chess strategist. Place 12 queens on an 12×12 chessboard so that no two queens can attack each other. Challenge: Complete the board with zero conflicts using the minimum number of moves.	Online N-Queens Game	https://www.n-queens.com
3	Water Jug Puzzle: You are an AI agent solving a water jug problem. You have two jugs with capacities of 11 litres and 9 litres. Neither jug has measurement markings. Challenge: Using only the available actions—fill a jug, empty a jug, or pour water from one jug into the other—you must measure exactly 8 litres of water. Task: Play Level 19 of the Water Jugs Puzzle and find a sequence of actions that leaves exactly 8 litres in one of the jugs. Try to solve the puzzle using the minimum number of moves.	Tic-Tac-Toe	https://playtictactoe.org
4	Connect Four AI Challenge: You are playing against an AI opponent. Your goal is to connect four of your pieces in a row before the AI does. Challenge: Plan your moves carefully and defeat the AI opponent.	Connect Four	https://www.mathsisfun.com/games/connect4.html

5	8-Puzzle AI Challenge: You are given a scrambled 8-puzzle. Move the tiles one at a time and arrange them in the correct order. Challenge: Solve the puzzle using the minimum possible number of moves.	8-Puzzle Game	https://8puzzle.org/
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Structure of the Report – Game-Based Learning

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of the Game-Based Activity	Provide the name of the game or puzzle and the level/challenge attempted.
2	Game Platform / Link	Mention the platform or website used and provide the corresponding game link.
3	Objective of the Activity	Explain the objective of the game and the target that needs to be achieved.
4	Problem Statement	Describe the problem or challenge given in the game. What is the student required to solve?
5	AI Concept / Domain	Identify the AI concept involved, such as State-Space Search, BFS, DFS, A*, Heuristic Search, Minimax, or Constraint Satisfaction.
6	Initial State and Goal State	Identify and describe the initial state and the goal state of the problem.
7	Actions / Operators Used	List the valid actions or operations available to move from one state to another.
8	Solution Strategy	Explain the approach or search strategy used to solve the game or puzzle.
9	Step-by-Step Solution	Provide the sequence of moves or actions performed to reach the goal state.
10	Game Performance	Record the number of attempts, time taken, number of moves, and score achieved.
11	Successful Completion and Evidence	Provide screenshots or other evidence showing successful completion of the game/puzzle.
12	Efficiency and Optimality	Analyse whether the solution was optimal or efficient. Can the puzzle be solved using fewer moves or less time?
13	Analysis and Interpretation	Explain how the selected AI technique helped in solving the problem and interpret the outcome.
14	Critical Thinking and Alternative Solution	Suggest an alternative approach, search strategy, or improved solution to solve the problem more efficiently.
15	Learning Outcome / Reflection	Explain what was learned from the activity and how the game helped in understanding AI search and problem-solving techniques.
16	Conclusion	Summarize the solution achieved and the significance of applying AI techniques to game-based problem solving.
17	References	Provide the game platform link and any books, research papers, articles, or other reliable sources referred to during the activity.

Evaluation Rubrics

Criteria	Percentage (%)	Excellent	Good	Satisfactory	Needs Improvement
Understanding of the Problem / Game Objective	10%	Clearly understands the problem and game objective	Understands the objective with minor errors	Partially understands the problem	Does not understand the objective
Application of AI Search / Problem-Solving Technique	15%	Correctly applies the appropriate AI search or problem-solving technique	Applies the technique with minor errors	Shows limited application of the technique	Unable to apply the appropriate technique
Successful Completion of the Game / Puzzle	20%	Successfully solves/completes the game independently	Completes the game with minimal assistance	Partially completes the challenge	Unable to complete the challenge
Efficiency of Solution / Optimal Moves	10%	Solves using optimal or highly efficient moves	Uses a reasonably efficient sequence of moves	Uses several unnecessary moves	Solution is highly inefficient or incomplete
Documentation / Evidence of Completion	15%	Provides clear steps, score, screenshots, and evidence of completion	Provides most required evidence	Provides incomplete steps or evidence	Provides no proper documentation or evidence
Analysis and Interpretation of the Solution	15%	Clearly analyses the solution and explains the AI technique and outcome	Provides a good analysis with minor gaps	Provides limited analysis or explanation	Unable to analyse or interpret the solution
Originality and Critical Thinking	15%	Demonstrates excellent independent thinking and explores an effective or alternative solution	Demonstrates good independent thinking	Shows limited critical thinking	Shows little or no independent thinking

Game-Based Learning Assessment Report

CO1- Assessment Tool 3

**Submitted in the partial fulfilment for the Course of
MLA0104-Artificial Intelligence and Expert System**

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted by

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1.AI Maze Escape Using Breadth-First Search (BFS)

1. Introduction:

The AI Maze Escape activity demonstrates how an intelligent agent can navigate through a maze by moving from the starting point to the goal while avoiding obstacles. The maze consists of open paths and blocked areas, requiring the agent to choose only valid movements. This activity introduces students to the concept of problem-solving through artificial intelligence.

The Breadth-First Search (BFS) algorithm is used to solve the maze because it explores all possible paths level by level. Since each movement has the same cost, BFS guarantees the shortest path to the destination. The activity was completed in Scratch, and an additional Python implementation was created to demonstrate the same AI concept automatically.

2. Objective of the Activity:

The main objective of this activity is to find the shortest path from the starting position to the goal without crossing obstacles. The activity helps students understand how search algorithms are applied to solve navigation problems in artificial intelligence.

Another objective is to improve logical thinking and problem-solving skills by analysing different paths before reaching the destination. The Python implementation further demonstrates how the BFS algorithm automatically finds the optimal solution.

3. AI Concept Used:

The AI concept used in this activity is Breadth-First Search (BFS), which is an uninformed search algorithm. BFS explores neighbouring nodes first before moving to deeper levels, ensuring that the shortest path is found when all moves have equal cost.

The algorithm uses a queue to store unexplored positions and a visited list to avoid revisiting the same location. BFS is widely used in robotics, path planning, navigation systems, and game development because of its reliability and optimality.

4. Initial State and Goal State:

The initial state is the position where the orange ball starts inside the maze. At this stage, the agent has not moved, and the objective is to reach the destination while avoiding all obstacles.

The goal state is reached when the orange ball touches the green goal area. After reaching the destination, the Scratch project displays the message "You win!", confirming that the maze

has been solved successfully.

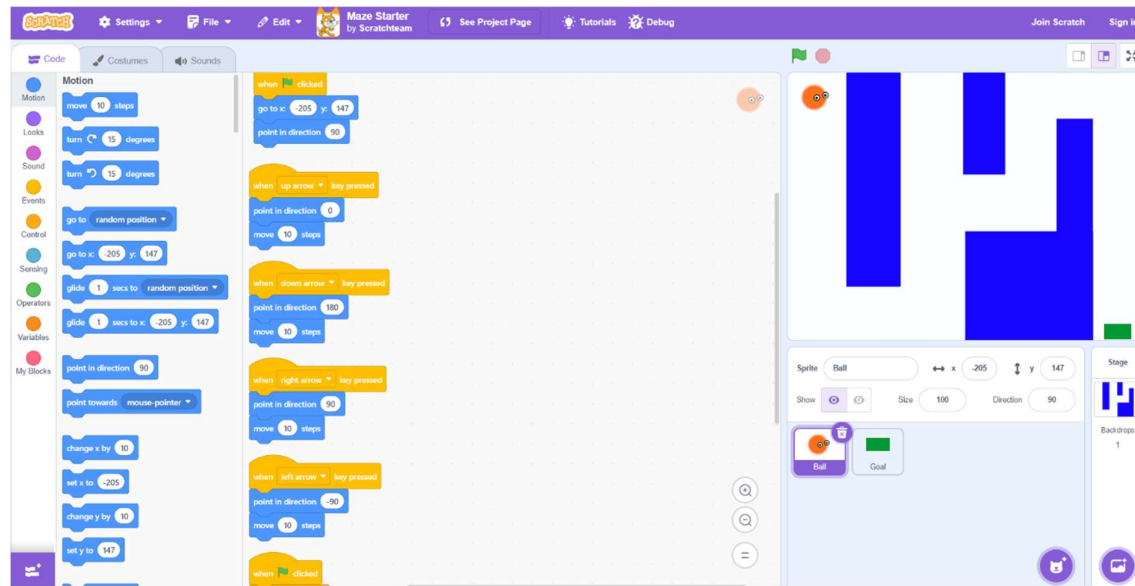


Figure 1: Initial state showing the starting position of the agent.

5. Actions / Operators:

The agent can perform four basic actions: Move Up, Move Down, Move Left, and Move Right. Each movement changes the current position of the agent within the maze.

A movement is considered valid only if the next position is inside the maze and is not blocked by a wall. These operators allow the agent to explore different paths until the destination is reached.

6. Solution Strategy:

The maze is solved using the Breadth-First Search algorithm. The search begins from the starting point and explores all neighbouring positions one level at a time until the goal is found.

Every visited position is stored to prevent repeated exploration. Once the destination is reached, the shortest path is reconstructed. This method guarantees an optimal solution for an unweighted maze.

7. Step-by-Step Solution:

The agent begins at the starting position and explores the maze by checking all valid neighbouring cells. Invalid moves that lead to walls or outside the maze are ignored.

The search continues until the goal is found. In the Python implementation, the shortest path is displayed along with the movement sequence and total number of steps required to reach the destination.

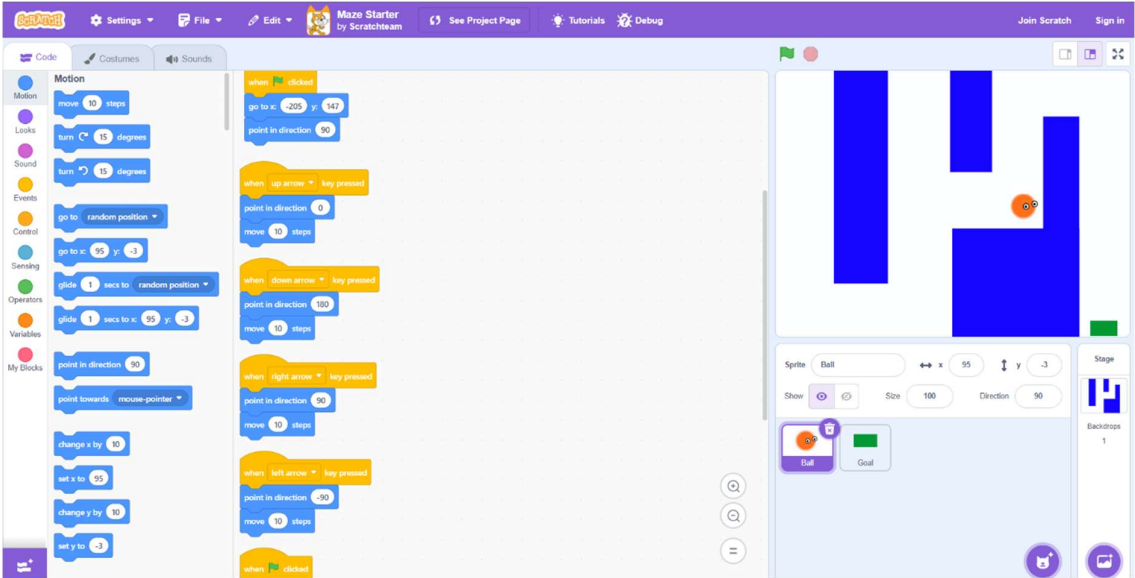


Figure 2: Agent moving through the maze using valid paths.

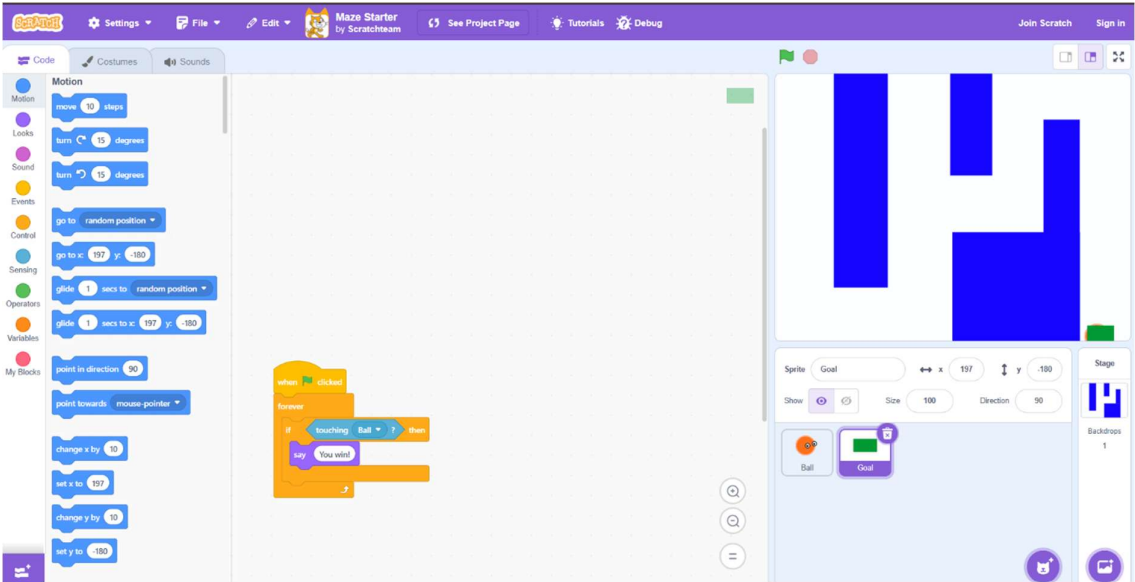


Figure 3: Agent successfully reaches the goal.

8. Game Performance:

The maze was completed successfully in Scratch, and the goal was reached without crossing any obstacles. The final screen displayed the "You win!" message, confirming successful completion.

The Python program also solved the maze automatically using BFS. It displayed the shortest path, movement sequence, and total number of moves, demonstrating the correctness of the algorithm.

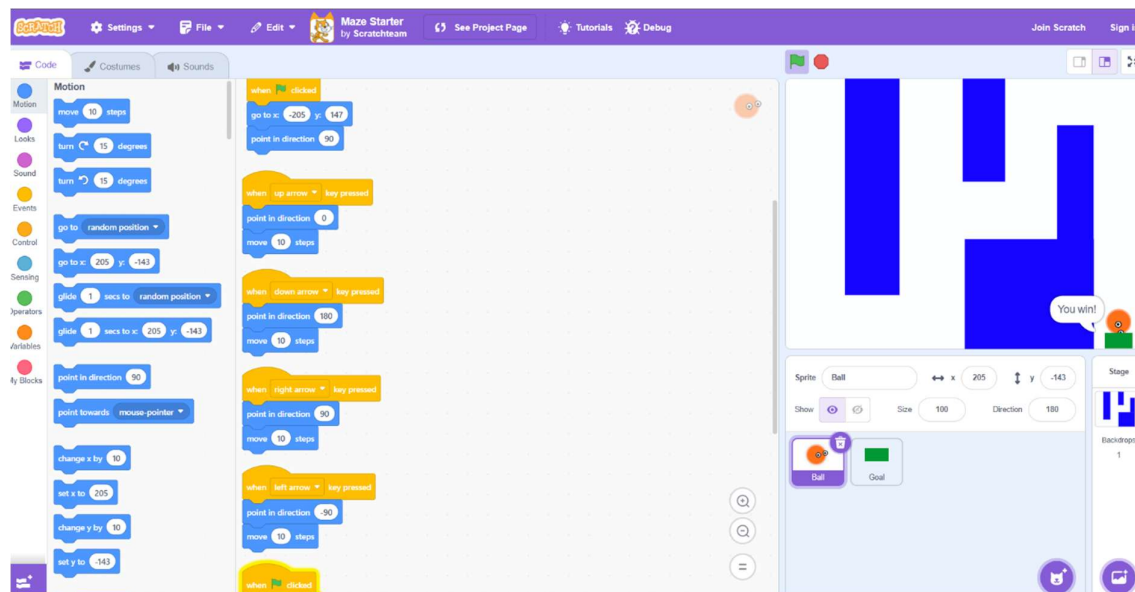


Figure 4: Final output displaying the "You win!" message.

9. Evidence / Output:

The activity is supported with screenshots showing the complete execution process. The first screenshot shows the initial state, followed by a screenshot during navigation, another showing the goal being reached, and the final "You win!" screen.

The report also includes the complete Python source code, Python IDLE output, graphical maze output, and the movement sequence generated by the BFS algorithm as additional evidence.

```
File Edit Shell Debug Options Window Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/rashm/OneDrive/Desktop/PROJECTS/ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEM/1/ai_maze_escape.py
Shortest path found using BFS
Path coordinates: [(0, 0), (0, 1), (0, 2), (1, 2), (2, 2), (2, 3), (2, 4), (3, 4), (4, 4), (4, 5), (4, 6), (5, 6), (6, 6)]
Movement sequence: RIGHT → RIGHT → DOWN → DOWN → RIGHT → RIGHT → DOWN → DOWN → RIGHT → RIGHT → DOWN → DOWN
Total number of moves: 12
```

Figure 5: Python Shell Output

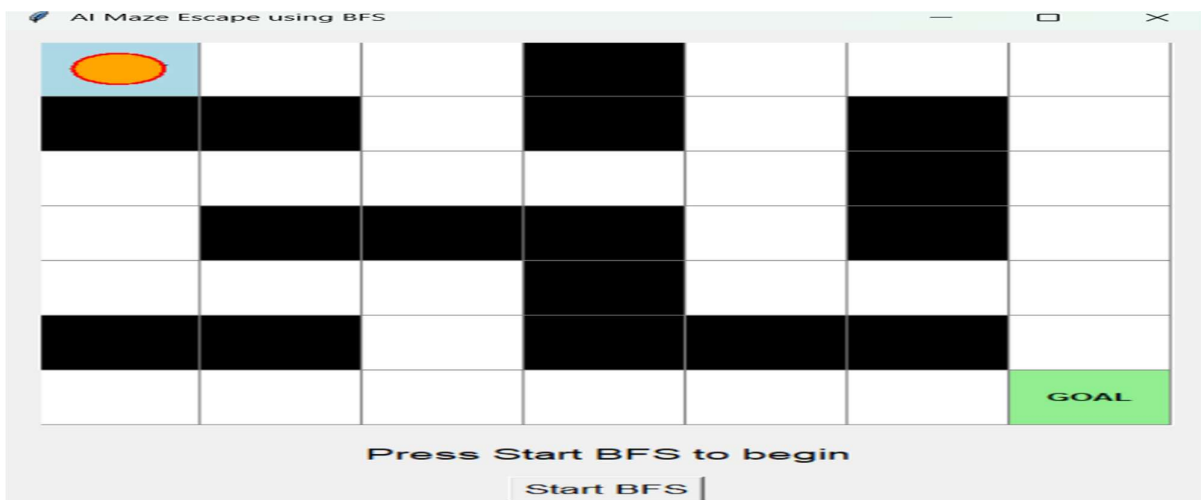


Figure 6: Python Maze Initial Window

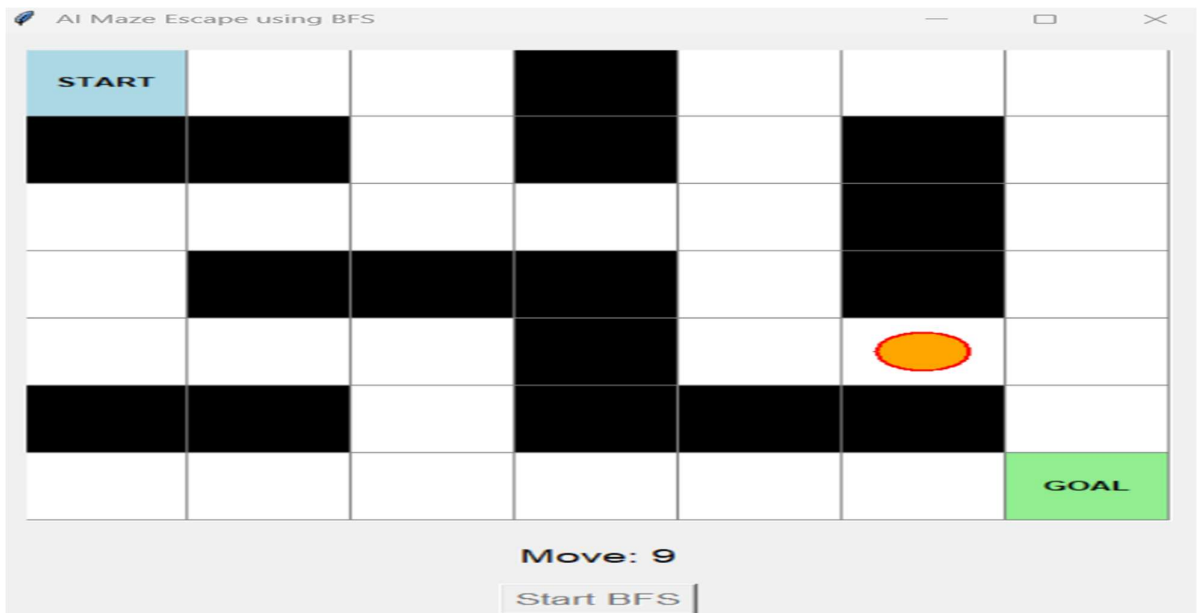


Figure 7: Python Maze During Execution

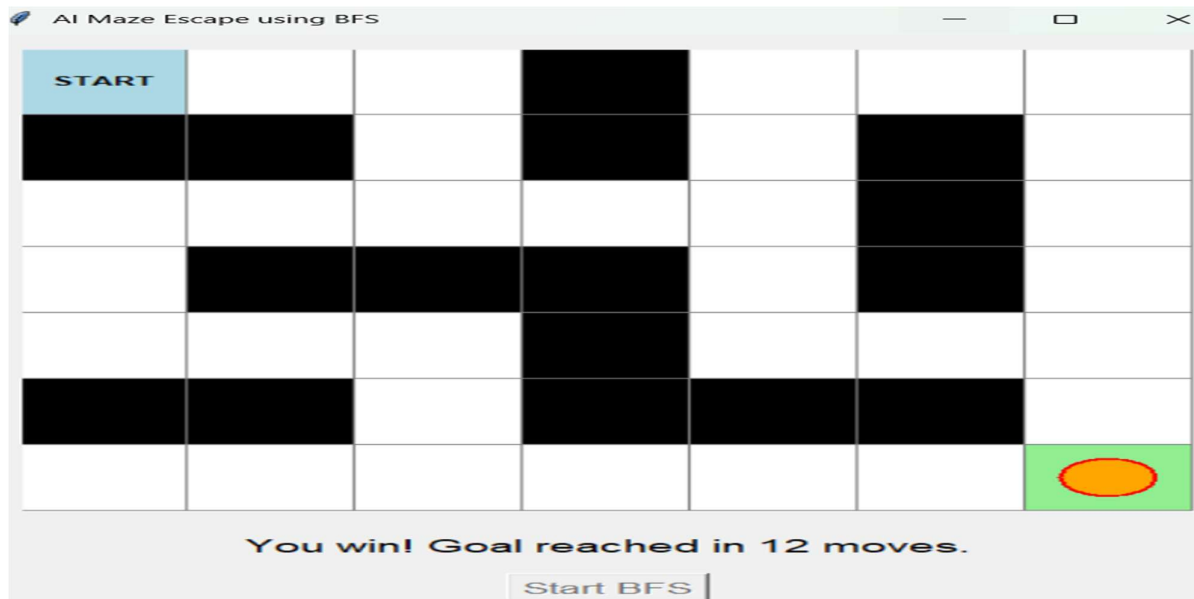


Figure 8: Python Final Output ("You win!")

10. Analysis:

Breadth-First Search successfully solved the maze by exploring all possible paths systematically. The algorithm avoided revisiting explored positions, making the search organised and efficient.

Since every movement had equal cost, BFS guaranteed the shortest path to the destination. This activity clearly demonstrated how AI search algorithms can solve navigation problems accurately.

11. Alternative Solution:

Another suitable algorithm for this problem is A* Search. Unlike BFS, A* uses a heuristic function to estimate the remaining distance to the goal, reducing the number of explored nodes.

For larger mazes, A* generally performs faster while still producing the shortest path when an appropriate heuristic is used. Therefore, it is commonly used in modern navigation and path-planning systems.

12. Learning Outcome / Reflection:

This activity helped me understand how artificial intelligence solves navigation problems using search algorithms. I learned how states, actions, and goal conditions are represented in a maze environment.

Developing the Python implementation improved my programming skills and gave me practical knowledge of the BFS algorithm. The activity also enhanced my logical thinking and problem-solving abilities.

13. Conclusion:

The AI Maze Escape activity was completed successfully using the Breadth-First Search algorithm. Both the Scratch implementation and the Python program correctly found a valid path from the start to the goal.

This activity provided practical experience in applying AI search techniques to real-world navigation problems. It improved my understanding of BFS, state-space search, and shortest-path algorithms through both manual and automated implementations.

14. References:

The Scratch platform was used to complete the game-based activity and demonstrate maze navigation. Python IDLE was used to implement the Breadth-First Search algorithm and generate the shortest path automatically.

The concepts used in this report are based on the Artificial Intelligence and Expert Systems (MLA0104) course material, particularly the topics related to state-space search and Breadth-First Search (BFS).

Question 2 – 12 Queens Challenge

1. 12-Queens Challenge Using Backtracking Algorithm (Constraint Satisfaction Problem):

The 12-Queens Challenge is an artificial intelligence puzzle in which twelve queens must be placed on a 12×12 chessboard without attacking each other. A queen can move horizontally, vertically, and diagonally, so no two queens should be placed in the same row, column, or diagonal. This activity demonstrates how AI solves complex problems by satisfying multiple constraints simultaneously.

The challenge is a classic Constraint Satisfaction Problem (CSP). In this activity, the puzzle was first solved using the online 12-Queens game and then implemented in Python using the Backtracking Algorithm. Both implementations demonstrate how AI systematically searches for a valid solution while following all the constraints.

2. Objective of the Activity:

The main objective of this activity is to place twelve queens on a 12×12 chessboard so that none of the queens attack each other. Every queen must occupy a unique row and column while also satisfying the diagonal constraints.

Another objective is to understand how artificial intelligence uses constraint satisfaction techniques to solve complex search problems. The Python implementation helps in understanding how the Backtracking Algorithm automatically finds a valid arrangement by checking constraints and backtracking whenever a conflict occurs.

3. AI Concept Used:

The AI concept used in this activity is the Constraint Satisfaction Problem (CSP). In a CSP, the objective is to find values for variables while satisfying all given constraints. Here, each queen represents a variable, and its position on the chessboard must satisfy the row, column, and diagonal constraints.

The Backtracking Algorithm is used to solve this problem. It places queens one by one and checks whether each placement is safe. If a conflict occurs, the algorithm removes the last queen and tries another position until a complete solution is found.

4. Initial State and Goal State:

The initial state is an empty 12×12 chessboard with no queens placed. At this stage, the objective is to place twelve queens without violating any constraints.

The goal state is achieved when all twelve queens are successfully placed on the board with no conflicts. The completed board represents a valid solution where every queen is safe from attack.

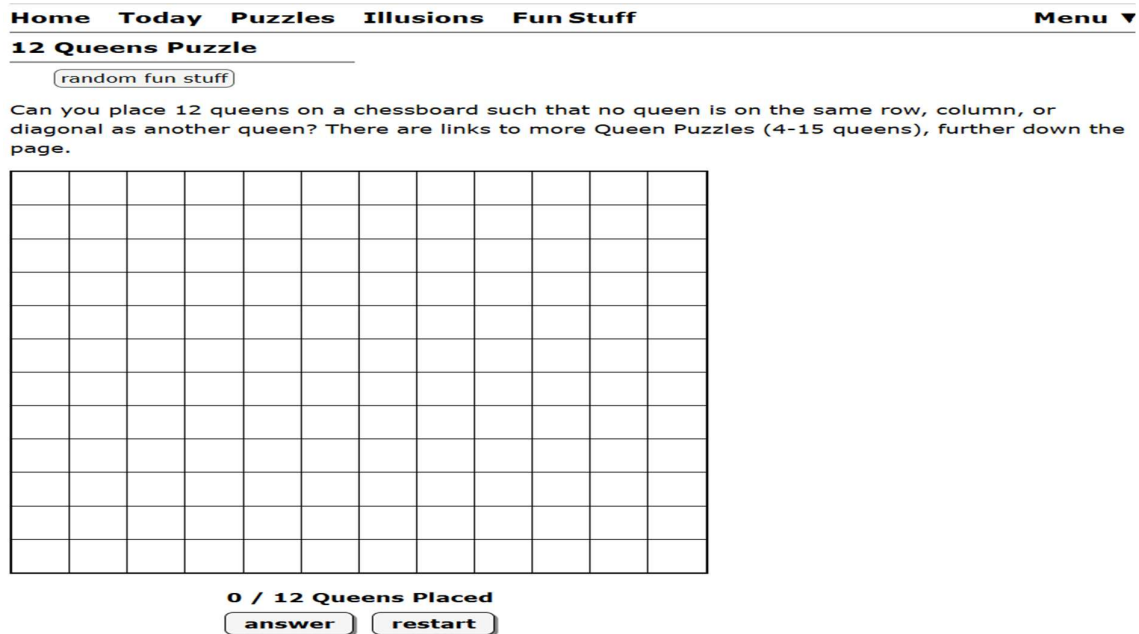


Figure 1: Empty 12×12 board

5. Actions / Operators:

The main action performed in this activity is placing a queen in a safe position on the chessboard. Before placing a queen, the algorithm checks whether the selected square satisfies all constraints. If a conflict is detected, the algorithm removes the previously placed queen and tries another position. This process continues until all twelve queens are placed successfully.

6. Solution Strategy:

The puzzle is solved using the Backtracking Algorithm. The algorithm begins by placing the first queen in a safe position. It then moves to the next column and repeats the process for the remaining queens.

Whenever no safe position is available, the algorithm returns to the previous queen, changes its position, and continues searching. This systematic approach ensures that all possible arrangements are explored until a valid solution is found.

7. Step-by-Step Solution:

The solution begins with an empty chessboard. Queens are placed one by one while checking row, column, and diagonal constraints. If a placement creates a conflict, the algorithm backtracks and selects another position.

This process continues until all twelve queens are safely placed on the board. The final arrangement satisfies all constraints and represents the correct solution to the puzzle.

12 Queens Puzzle

random fun stuff

Can you place 12 queens on a chessboard such that no queen is on the same row, column, or diagonal as another queen? There are links to more Queen Puzzles (4-15 queens), further down the page.

•	♛	•	•	•	•	•	•	•	•	•	•
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•	•	•		•				•	•		•
	•			•				•	•		

4 / 12 Queens Placed

answer

restart

Figure 2: Board after placing all 12 queens

8. Game Performance:

The online 12-Queens puzzle was completed successfully by placing all twelve queens without conflicts. The final board satisfied every constraint required by the game.

The Python implementation also produced a valid solution using the Backtracking Algorithm. The output displayed the complete arrangement of queens on the chessboard, confirming the correctness of the algorithm.

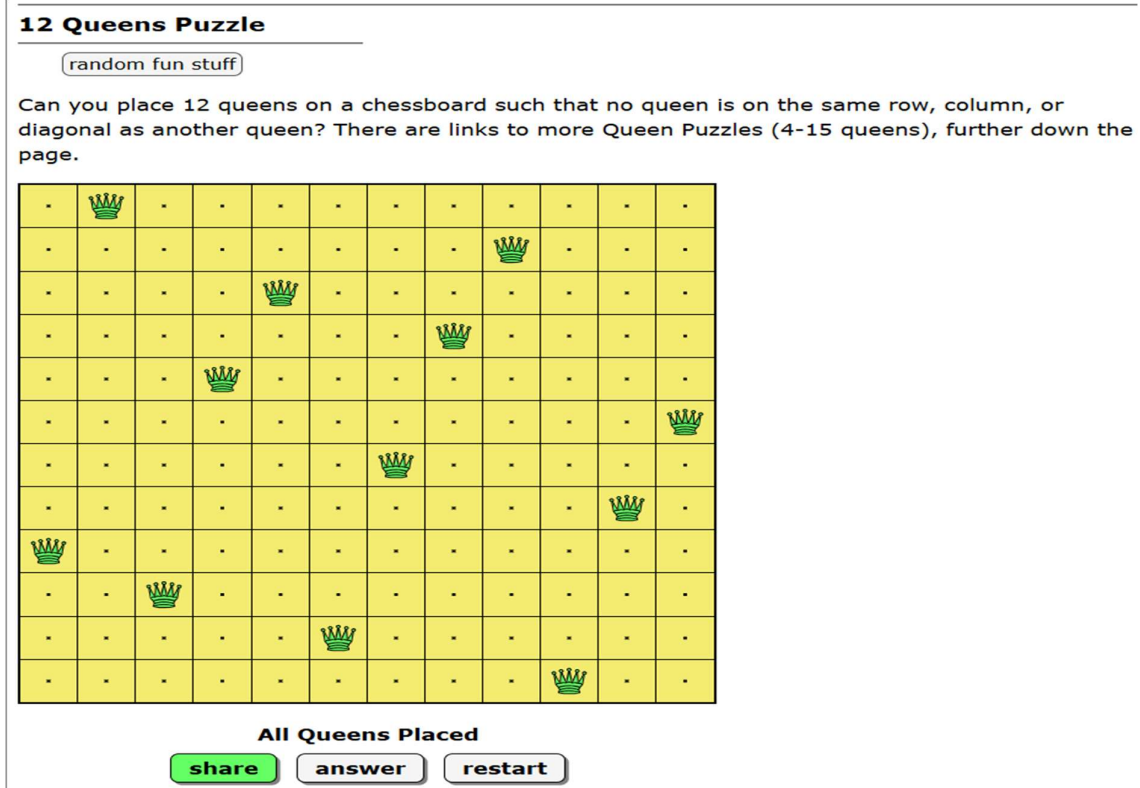


Figure 3: Final solved board

9. Evidence / Output:

The report includes screenshots of the complete solving process. These screenshots show the initial empty board, the intermediate stage, and the final solved board from the website. Additional evidence includes the Python source code, Python IDLE output, and the final solution generated by the Backtracking Algorithm.

By implementing the solution in Python, I also improved my programming and problem-solving skills. The activity enhanced my understanding of Constraint Satisfaction Problems and their applications in AI.

13. Conclusion:

The 12-Queens Challenge was successfully completed using the online puzzle and the Python Backtracking implementation. Both methods produced a valid arrangement in which no two queens attacked each other.

This activity provided practical knowledge of Constraint Satisfaction Problems, Backtracking, and AI-based search techniques. It demonstrated how artificial intelligence can solve complex logical problems efficiently.

14. References:

1. Artificial Intelligence and Expert Systems (MLA0104) Course Material.
2. 12-Queens Online Puzzle Website.
3. Python 3 Documentation – <https://docs.python.org/3/>

Question 3 – Water Jug Puzzle

1. Water Jug Puzzle Using State Space Search:

The Water Jug Puzzle is a classic Artificial Intelligence problem in which two water jugs of different capacities are used to measure an exact quantity of water. In this assessment, the objective is to obtain exactly 8 litres using an 11-litre jug and a 9-litre jug. Only three operations are allowed: filling a jug, emptying a jug, and pouring water from one jug to the other.

This activity demonstrates how AI solves a problem by exploring different states until the goal state is reached. Each combination of water levels in the two jugs represents a state, and the solution is obtained by performing valid actions in a systematic sequence.

2. Objective of the Activity:

The objective of this activity is to measure exactly 8 litres of water using the given jugs while following the puzzle rules. The solution should be achieved using a logical sequence of operations without violating the constraints.

This activity helps students understand how AI search algorithms solve state-space problems by exploring different possibilities. It also develops logical thinking, planning, and problem-solving skills.

3. AI Concept Used:

The AI concept used in this activity is State Space Search. Every possible amount of water in the two jugs represents a different state. The search starts from the initial state and continues until the required goal state is reached.

The solution can be obtained using search algorithms such as Breadth-First Search (BFS) or Depth-First Search (DFS). BFS is commonly preferred because it can find the minimum number of moves required to reach the target state.

4. Initial State and Goal State:

The initial state consists of both jugs being empty. At this stage, no water has been poured into either jug, and the search process has not yet started.

The goal state is reached when exactly 8 litres of water are present in one of the jugs. Once this state is achieved, the puzzle is considered solved successfully.

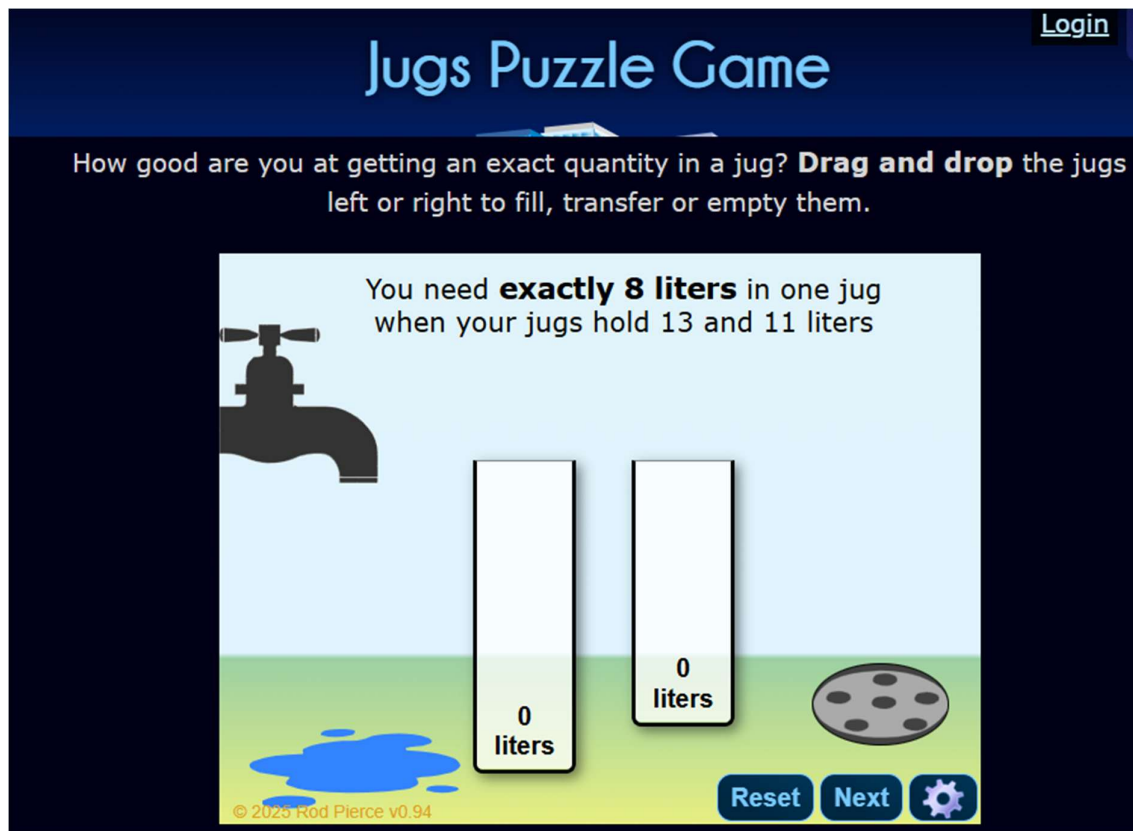


Figure 1: Initial state of the Water Jug Puzzle.

5. Actions / Operators:

The puzzle allows only three basic operations: Fill, Empty, and Pour. A jug can be filled completely, emptied completely, or water can be transferred from one jug to another until either the source is empty or the destination is full.

Each operation changes the current state of the puzzle. By applying these operators in the correct order, the required quantity of water can be measured successfully.

6. Solution Strategy:

The puzzle is solved by representing every water level combination as a state and moving systematically from one state to another. Invalid or repeated states are avoided while searching for the goal.

The sequence of operations continues until the required quantity of **8 litres** is obtained. The final state represents the successful completion of the puzzle.

7. Step-by-Step Solution:

The solution starts with both jugs empty. The larger jug is filled, water is transferred between the jugs, and the jugs are filled or emptied whenever necessary. Each operation changes the current state.

The process continues until exactly 8 litres of water remain in one of the jugs. The final state satisfies the required condition and completes the puzzle successfully.



Figure 2: Intermediate stage of the puzzle.

8. Game Performance:

The Water Jug Puzzle was completed successfully by obtaining exactly **8 litres** using the allowed operations. All actions followed the puzzle rules, and the goal state was achieved.

The Python implementation also produced the required result automatically by exploring the valid states. This confirmed the correctness of the search algorithm used to solve the problem.



Figure 3: Final state showing exactly 8 litres.

9. Evidence / Output:

The report includes screenshots of the initial state, intermediate stages, and the final successful solution from the game. These screenshots clearly show the progress of the activity.

The Python implementation is also included with the complete source code, Python IDLE output, and the final state generated by the program. These outputs provide additional evidence of the successful solution.

```

water_jug_bfs.py - C:/Users/rashm/OneDrive/Desktop/PROJECTS/ARTIFICIAL INTELLIGENCE...
File Edit Format Run Options Window Help
from collections import deque

CAPACITY_A = 11
CAPACITY_B = 9
TARGET = 8

def get_next_states(state):
    """Return all valid states reachable from the current state."""
    a, b = state
    next_states = []

    # Fill either jug
    next_states.append(((CAPACITY_A, b), "Fill the 11L jug"))
    next_states.append(((a, CAPACITY_B), "Fill the 9L jug"))

    # Empty either jug
    next_states.append(((0, b), "Empty the 11L jug"))
    next_states.append(((a, 0), "Empty the 9L jug"))

    # Pour from 11L jug to 9L jug
    transfer = min(a, CAPACITY_B - b)
    next_states.append(
        ((a - transfer, b + transfer), "Pour 11L jug into 9L jug")
    )

    # Pour from 9L jug to 11L jug
    transfer = min(b, CAPACITY_A - a)
    next_states.append(
        ((a + transfer, b - transfer), "Pour 9L jug into 11L jug")
    )

    return next_states

def bfs():
    """Find the shortest sequence of operations using BFS."""
    initial_state = (0, 0)
    queue = deque([(initial_state, [])])
    visited = {initial_state}

```

Figure 4: Python source code.


```
IDLE Shell 3.12.5
File Edit Shell Debug Options Window Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/rashm/OneDrive/Desktop/PROJECTS/ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEM/1/water_jug_bfs.py
WATER JUG PUZZLE USING BFS
-----
Jug A Capacity : 11 litres
Jug B Capacity : 9 litres
Target          : 8 litres
-----
Initial State : (0, 0)
Step 1: Fill the 11L jug
State = (11, 0)
Step 2: Pour 11L jug into 9L jug
State = (2, 9)
Step 3: Empty the 9L jug
State = (2, 0)
Step 4: Pour 11L jug into 9L jug
State = (0, 2)
Step 5: Fill the 11L jug
State = (11, 2)
Step 6: Pour 11L jug into 9L jug
State = (4, 9)
Step 7: Empty the 9L jug
State = (4, 0)
Step 8: Pour 11L jug into 9L jug
State = (0, 4)
Step 9: Fill the 11L jug
State = (11, 4)
Step 10: Pour 11L jug into 9L jug
State = (6, 9)
Step 11: Empty the 9L jug
State = (6, 0)
Step 12: Pour 11L jug into 9L jug
State = (0, 6)
Step 13: Fill the 11L jug
State = (11, 6)
Step 14: Pour 11L jug into 9L jug
State = (8, 9)
-----
Goal reached at state: (8, 9)
Number of moves: 14
>>>
```

Figure 5: Python output.

10. Analysis:

The Water Jug Puzzle demonstrates how Artificial Intelligence solves state-space problems by exploring different possible states. Instead of guessing randomly, the algorithm follows a logical sequence of actions until the goal state is reached.

This activity shows the importance of search strategies in AI. By representing the problem as states and operators, the puzzle can be solved efficiently while avoiding unnecessary actions.

11. Alternative Solution:

Another approach to solving the Water Jug Puzzle is Depth-First Search (DFS). DFS explores one possible path completely before trying another path. Although it can find a solution, it may not always produce the minimum number of moves.

Compared with DFS, Breadth-First Search (BFS) is generally preferred for this problem because it guarantees the shortest solution when each operation has the same cost.

12. Learning Outcome / Reflection:

This activity helped me understand how Artificial Intelligence represents problems using states and operators. I learned how different search algorithms can be applied to find a valid solution while following the given constraints.

By solving the puzzle manually and implementing it in Python, I improved my logical thinking, programming skills, and understanding of state-space search techniques used in AI applications.

13. Conclusion:

The Water Jug Puzzle was successfully completed by measuring exactly **8 litres** using the 11-litre and 9-litre jugs. The activity demonstrated the practical application of state-space search in solving real-world problems.

Both the game-based activity and the Python implementation showed how AI search techniques can systematically reach the goal state. This activity improved my understanding of AI problem-solving, search algorithms, and logical reasoning.

14. References:

1. Artificial Intelligence and Expert Systems (MLA0104) Course Material.
2. Water Jug Puzzle (Game platform specified in the assessment).
3. Python 3 Documentation – <https://docs.python.org/3/>

Question 4 – Connect Four AI Challenge

1. Connect Four AI Challenge Using Minimax Algorithm:

The Connect Four AI Challenge is a strategy-based game where the player competes against an Artificial Intelligence opponent. The objective is to connect four pieces in a horizontal, vertical, or diagonal line before the AI achieves the same goal. Every move affects the future state of the game, making planning and decision-making important.

This activity demonstrates how Artificial Intelligence can be applied to two-player games. The AI opponent predicts possible future moves and selects the best action using game-search techniques. By playing against the AI, students learn how intelligent decision-making algorithms are used in competitive environments.

2. Objective of the Activity:

The objective of this activity is to defeat the AI opponent by connecting four consecutive pieces before the AI completes its own sequence. The player must carefully choose each move while blocking the opponent's winning opportunities.

The activity also helps students understand strategic thinking and decision-making in Artificial Intelligence. It demonstrates how search algorithms evaluate different game states to determine the best possible move.

3. AI Concept Used:

The AI concept used in this activity is the Minimax Algorithm. Minimax is an adversarial search algorithm used in two-player games where one player tries to maximize the score while the opponent tries to minimize it.

The algorithm evaluates possible future game states before selecting a move. It predicts the opponent's responses and chooses the action that provides the highest chance of winning while reducing the opponent's advantage.

4. Initial State and Goal State:

The initial state is an empty Connect Four board with no pieces placed by either player. The game begins with the player making the first move or following the game's turn order.

The goal state is achieved when the player successfully connects four pieces in a row horizontally, vertically, or diagonally before the AI opponent. If the AI connects four pieces first, the player loses the game.



Figure 1: Initial game board.

5. Actions / Operators:

The player can place one piece into any valid column on the board. The piece automatically falls to the lowest available position in that column.

Each move changes the game state and creates new possibilities for both players. The player must also block the AI whenever it is close to forming four connected pieces.

6. Solution Strategy:

The strategy is to create multiple opportunities to connect four pieces while preventing the AI from completing its own sequence. Every move is planned by considering future board positions instead of making random decisions.

The AI uses the Minimax algorithm to evaluate possible moves. The player must therefore think ahead and select moves that maximize winning chances while minimizing the AI's opportunities.

7. Step-by-Step Solution:

The game begins with an empty board. The player places the first piece in a strategic position, usually near the center columns, to increase future winning possibilities. The AI responds with its own move. The game continues until one player connects four consecutive pieces. Careful planning, blocking the AI, and creating multiple winning opportunities help achieve victory.



Figure 2: Board during gameplay.

8. Game Performance:

The Connect Four game was completed successfully by following a planned strategy. The player continuously blocked the AI's attacks while creating opportunities to connect four pieces.

The final result showed a successful win against the AI opponent. The activity demonstrated effective strategic thinking and proper application of AI game-search concepts.



Figure 3: Final winning board.

9. Evidence / Output:

The report includes screenshots showing the complete gameplay process, including the initial board, intermediate game states, and the final winning position. These screenshots provide evidence of successful completion.

The Python implementation using the Minimax algorithm can also be included with the source code and output screenshots. This demonstrates how AI automatically evaluates the best moves during gameplay.

```

connect_four_ai.py - C:/Users/rashmy/OneDrive/Desktop/PROJECTS/ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEM/1/connect_four_ai.py (3.12.5)
File Edit Format Run Options Window Help
import tkinter as tk
from tkinter import messagebox
import math
import random

ROWS = 6
COLUMNS = 7
CELL_SIZE = 75

EMPTY = 0
HUMAN = 1
AI = 2

class ConnectFourAI:
    def __init__(self, root):
        self.root = root
        self.root.title("Connect Four AI Challenge")

        self.board = [
            [EMPTY for _ in range(COLUMNS)]
            for _ in range(ROWS)
        ]
        self.game_over = False

        tk.Label(
            root,
            text="CONNECT FOUR - HUMAN VS AI",
            font=("Arial", 20, "bold")
        ).pack(pady=10)

        self.status_label = tk.Label(
            root,
            text="Your Turn - Click Any Column",
            font=("Arial", 14, "bold")
        )
        self.status_label.pack(pady=5)

        self.canvas = tk.Canvas(
            root,
            width=COLUMNS * CELL_SIZE,
            height=ROWS * CELL_SIZE,
            bg="blue"
        )
        self.canvas.pack(padx=20, pady=10)

        self.canvas.bind("<Button-1>", self.human_move)

        tk.Button(
            root,
            text="Reset Game",

```

Figure 4: Python code.

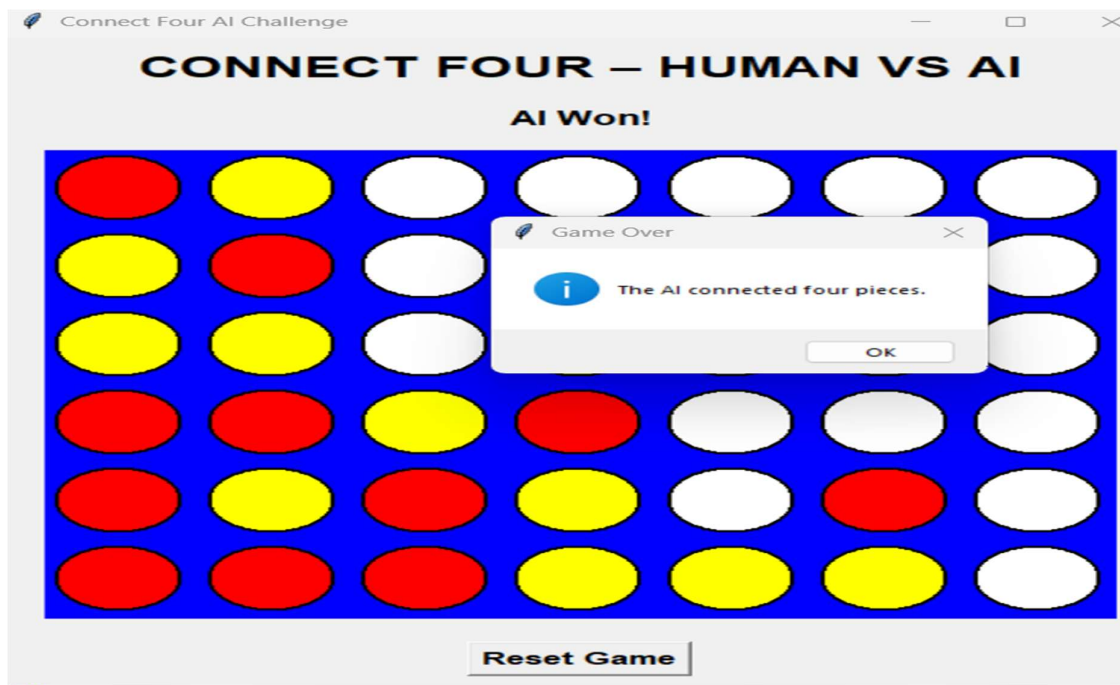


Figure 5: Python output.

10. Analysis:

The Connect Four game illustrates how Artificial Intelligence uses decision-making algorithms to compete against human players. Instead of choosing random moves, the AI evaluates future

possibilities before selecting an action. The Minimax algorithm improves game performance by considering both the player's and the opponent's moves. This makes the AI more intelligent and capable of making strategic decisions.

11. Alternative Solution:

An improved version of the Minimax algorithm is Minimax with Alpha-Beta Pruning. Alpha-Beta Pruning eliminates unnecessary branches during the search process, reducing computation time while still producing the same optimal move. Compared with the standard Minimax algorithm, Alpha-Beta Pruning is faster and more efficient, especially for games with a large search space such as Connect Four.

12. Learning Outcome / Reflection:

This activity helped me understand how Artificial Intelligence solves competitive games using decision-making algorithms. I learned how every move changes the game state and influences future decisions. By playing against the AI and analysing its moves, I improved my strategic thinking and gained practical knowledge of Minimax and adversarial search techniques used in AI.

13. Conclusion:

The Connect Four AI Challenge was successfully completed by applying strategic planning and understanding AI decision-making techniques. The activity demonstrated how game-search algorithms help an AI opponent choose intelligent moves. This activity improved my understanding of Minimax, adversarial search, and game strategy. It also showed how AI can be applied to solve competitive game problems through logical reasoning and future state evaluation.

14. References:

1. Connect Four Game – <https://www.mathsisfun.com/games/connect4.html> (as specified in the assessment)
2. Artificial Intelligence and Expert Systems (MLA0104) Course Material.
3. Python 3 Documentation – <https://docs.python.org/3/>

Question 5 – 8-Puzzle AI Challenge

1. 8-Puzzle AI Challenge Using A * Search Algorithm:

The 8-Puzzle AI Challenge is a classic Artificial Intelligence problem in which numbered tiles are arranged on a 3×3 board with one empty space. The objective is to move the tiles one at a time until they are arranged in the correct order. The puzzle requires careful planning because every move changes the current state of the board.

This activity demonstrates how AI search algorithms solve state-space problems efficiently. The puzzle was solved using an online 8-Puzzle game, and a Python implementation can be used to demonstrate the same AI concept automatically. The activity helps students understand heuristic search and optimal path finding.

2. Objective of the Activity:

The objective of this activity is to rearrange the scrambled tiles into the correct numerical order using the minimum possible number of moves. Only one tile adjacent to the empty space can be moved at a time.

The activity also helps students understand how Artificial Intelligence applies search algorithms to solve complex puzzles. It improves logical thinking, planning, and problem-solving skills while demonstrating optimal search techniques.

3. AI Concept Used:

The AI concept used in this activity is Heuristic Search, specifically the *A Search Algorithm**. A* combines the actual cost of reaching a state with a heuristic estimate of the remaining distance to the goal, making it efficient for solving puzzles.

The algorithm evaluates different board configurations and always selects the most promising state for exploration. Because of this heuristic guidance, A* usually finds the optimal solution with fewer explored states than uninformed search methods.

4. Initial State and Goal State:

The initial state is the scrambled arrangement of the numbered tiles displayed at the beginning of the game. The empty space allows the neighbouring tiles to move into it, creating new board configurations.

The goal state is achieved when the tiles are arranged in the correct order from **1 to 8**, with the empty space in the final position. Once this arrangement is reached, the puzzle is successfully solved.

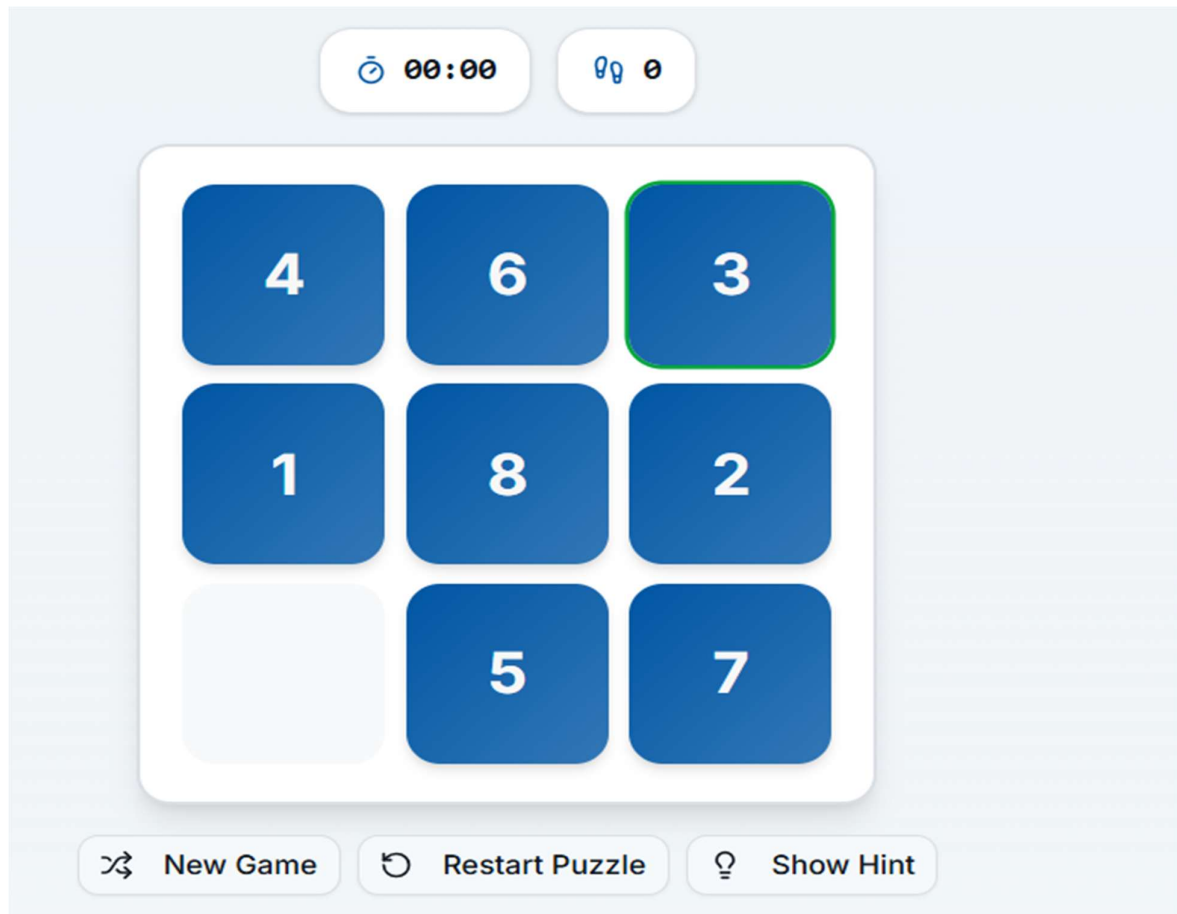


Figure 1: Initial scrambled puzzle.

5. Actions / Operators:

The puzzle allows the player to move one tile at a time into the adjacent empty space. Valid movements include moving a tile up, down, left, or right, depending on the position of the empty space.

Each move changes the current state of the puzzle. Invalid moves, such as moving non-adjacent tiles, are not allowed. The sequence of valid moves continues until the goal state is reached.

6. Solution Strategy:

The puzzle is solved by exploring possible board configurations while selecting the most promising state at each step. The search continues until the tiles are arranged in the correct order.

The A* Search algorithm uses a heuristic function to estimate the remaining distance to the goal. This reduces unnecessary exploration and helps the algorithm find an optimal solution efficiently.

7. Step-by-Step Solution:

The solution begins with the scrambled board. The empty space is used to move adjacent tiles into their correct positions while gradually reducing the number of misplaced tiles.

Each move brings the board closer to the goal state until all tiles are arranged in the correct order. The final configuration represents the successful completion of the puzzle.



Figure 2: Intermediate stage.

8. Game Performance:

The 8-Puzzle was completed successfully by arranging all tiles in the correct order. The solution was achieved using a logical sequence of moves while minimizing unnecessary movements.

The Python implementation also solved the puzzle successfully using the A* Search algorithm. The output displayed the sequence of board states from the initial arrangement to the final solution.

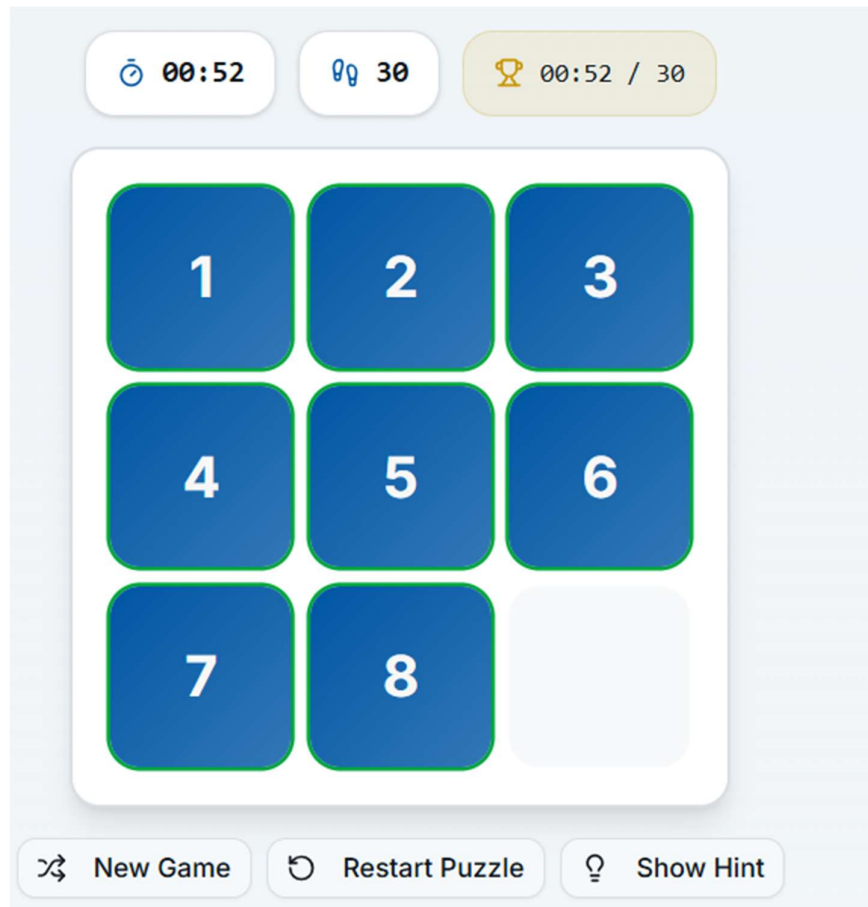


Figure 3: Final solved puzzle.

9. Evidence / Output:

The report includes screenshots showing the initial scrambled puzzle, intermediate moves, and the final solved board. These screenshots provide clear evidence of successful completion.

Additional evidence includes the Python source code, Python output, and the sequence of moves generated by the A* Search algorithm.

```

eight_puzzle_astar.py - C:/Users/rashm/OneDrive/Desktop/PROJECTS/ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEM/5/eight_puzzle_astar.py
File Edit Format Run Options Window Help
from queue import PriorityQueue

GOAL = (1, 2, 3,
        4, 5, 6,
        7, 8, 0)

MOVES = {
    0: [1, 3],
    1: [0, 2, 4],
    2: [1, 5],
    3: [0, 4, 6],
    4: [1, 3, 5, 7],
    5: [2, 4, 8],
    6: [3, 7],
    7: [4, 6, 8],
    8: [5, 7]
}

def heuristic(state):
    """Number of misplaced tiles."""
    count = 0
    for i in range(9):
        if state[i] != 0 and state[i] != GOAL[i]:
            count += 1
    return count

def neighbours(state):
    blank = state.index(0)
    result = []

    for move in MOVES[blank]:
        temp = list(state)
        temp[blank], temp[move] = temp[move], temp[blank]
        result.append(tuple(temp))

    return result

def print_board(state):
    for i in range(0, 9, 3):
        row = state[i:i + 3]
        print(" ".join("_" if x == 0 else str(x) for x in row))
    print()

def astar(start):
    pq = PriorityQueue()
    pq.put((heuristic(start), 0, start))

```

Figure 4: Python source code.

```

IDLE Shell 3.12.5
File Edit Shell Debug Options Window Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit
AMD64] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/rashm/OneDrive/Desktop/PROJECTS/ARTIFICIAL INTELLIGENCE AND
EXPERT SYSTEM/5/eight_puzzle_astar.py
INITIAL STATE

1 2 3
4 _ 6
7 5 8

A* SOLUTION FOUND

Step 0
1 2 3
4 _ 6
7 5 8

Step 1
1 2 3
4 5 6
7 _ 8
|
Step 2
1 2 3
4 5 6
7 8 _

Total Moves = 2
>>>

```

Figure 5: Python output.

10. Analysis:

The A* Search algorithm efficiently solved the puzzle by evaluating the most promising board configurations first. The heuristic function reduced the number of unnecessary states explored, making the search faster than uninformed methods.

This activity demonstrates how Artificial Intelligence uses heuristic search to solve optimization problems efficiently. It also highlights the importance of choosing an appropriate search strategy for state-space problems.

11. Alternative Solution:

An alternative approach is Breadth-First Search (BFS). BFS guarantees the shortest solution when all moves have equal cost, but it explores many more states and therefore requires more memory.

Compared with BFS, the A* algorithm is more efficient because it uses heuristic information to guide the search. This makes A* the preferred method for solving the 8-Puzzle efficiently.

12. Learning Outcome / Reflection:

This activity helped me understand how Artificial Intelligence solves state-space problems using heuristic search techniques. I learned how different board configurations represent different states and how search algorithms systematically explore them. By solving the puzzle manually and analysing the A* algorithm, I improved my logical reasoning, programming skills, and understanding of heuristic search in Artificial Intelligence.

13. Conclusion:

The 8-Puzzle AI Challenge was completed successfully by arranging the tiles into the correct order using a systematic search strategy. The activity demonstrated the practical application of heuristic search in solving optimization problems. This exercise improved my understanding of state-space search, A* Search, and AI problem-solving techniques. It also showed how heuristic algorithms can efficiently solve puzzles while minimizing unnecessary moves.

14. References:

1. 8-Puzzle Game: <https://8puzzle.org/>
2. Artificial Intelligence and Expert Systems (MLA0104) Course Material.
3. Python 3 Documentation – <https://docs.python.org/3/>